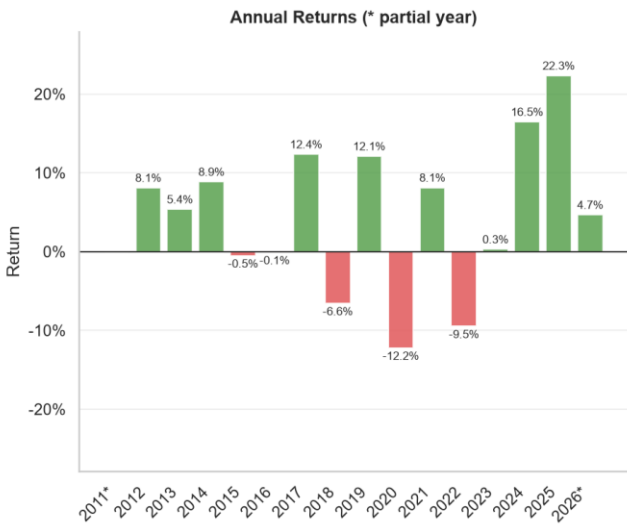
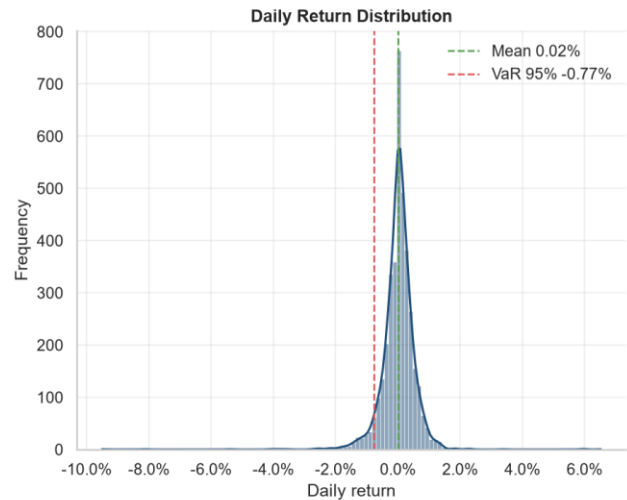
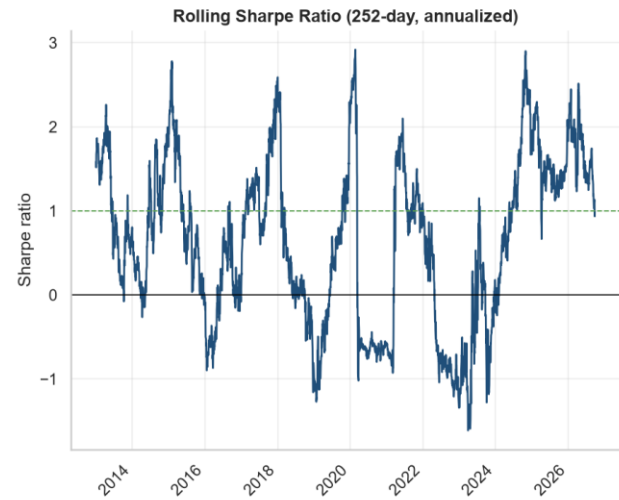
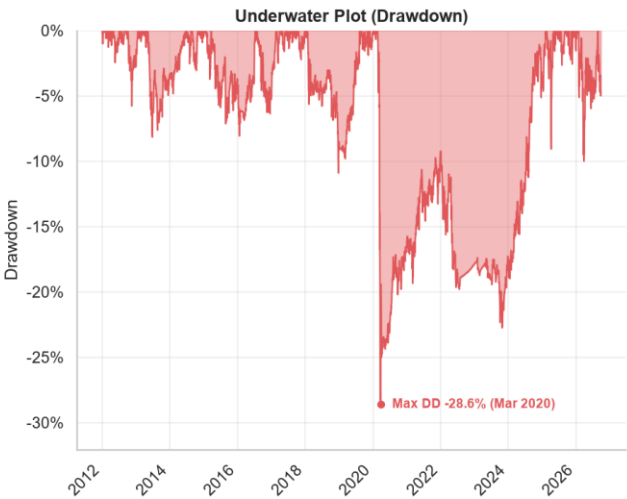
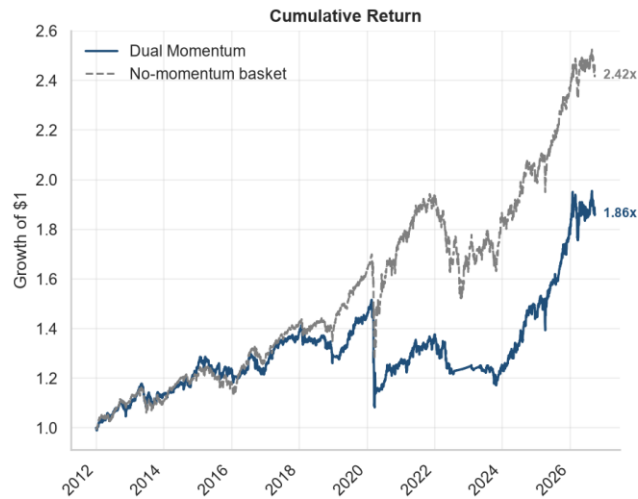


Antonacci (2016) Dual Momentum — Out-of-Sample Backtest

Dec 2011 – Sep 2026 · 14.7 years

	CAGR	Volatility	Sharpe (rf = 0)	Max Drawdown	Calmar	Longest DD
Dual Momentum	4.32%	8.96%	0.52	-28.62%	0.15	1249 sessions
No-momentum basket	6.19%	9.00%	0.71	-24.89%	0.25	667 sessions



Monthly Returns (%)

2011												0.0	0.0
2012	2.2	0.9	0.5	1.8	-0.4	0.7	2.0	1.1	1.0	-1.6	0.1	-0.3	8.1
2013	2.6	1.3	2.5	2.2	-4.9	-1.4	2.2	-2.8	0.9	2.7	-0.6	0.9	5.4
2014	-0.9	1.7	0.4	1.0	0.8	0.9	-2.5	4.0	-2.9	3.5	1.9	0.8	8.9
2015	3.3	-1.0	0.2	-1.8	-0.3	-2.8	3.1	-3.0	0.4	1.7	-0.3	0.1	-0.5
2016	-2.0	-0.2	0.3	1.6	-0.5	3.5	2.5	-2.0	-0.2	-2.8	-0.8	0.7	-0.1
2017	2.4	3.6	0.6	1.8	0.1	0.3	0.9	0.5	0.2	-0.3	0.1	1.7	12.4
2018	0.4	-2.8	0.0	0.1	-0.7	-0.4	1.4	1.1	-0.2	-3.0	1.2	-3.6	-6.6
2019	-0.5	-0.4	2.7	0.7	0.4	2.9	0.8	3.3	0.3	0.9	0.4	-0.1	12.1
2020	1.9	-2.2	-19.2	-1.0	0.2	1.6	4.6	1.6	-2.0	-0.8	1.8	2.8	-12.2
2021	-1.2	-1.3	1.8	3.6	2.1	-1.3	0.2	1.6	-1.8	2.6	-2.0	3.9	8.1
2022	-3.7	-1.7	2.5	-3.8	-2.3	-1.7	-0.4	0.3	0.2	0.2	0.3	0.4	-9.5
2023	0.3	-1.1	0.6	0.3	-1.2	-0.1	1.8	-0.6	-2.8	-2.0	3.0	2.4	0.3
2024	-0.8	1.8	3.6	-2.0	2.8	1.2	3.9	3.1	2.9	-0.1	2.2	-3.0	16.5
2025	3.0	1.3	-0.3	0.7	2.3	1.9	-0.7	2.4	4.1	1.6	2.9	1.3	22.3
2026	5.1	3.7	-6.9	4.1	0.8	-2.0	0.5	2.5	-2.6				4.7
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year

Net of 5 bps transaction costs · Total return (dividends reinvested) · Signal at month-end close, execution at next session's open · Benchmark: equal-weight 9-asset basket (8 risky assets + T-bills) without momentum · Data: FactSet

Dual Momentum, out of sample

Antonacci (2012, rev. 2016), replicated with ETFs from December 2011 to September 2026 (14.7 years)

The original strategy

In *Risk Premia Harvesting Through Dual Momentum*, Gary Antonacci builds a long-only portfolio split equally across four modules, each targeting a different risk premium: Equities (US vs. international stocks), Credit (high yield vs. intermediate credit), REITs (equity vs. mortgage REITs) and Economic Stress (long Treasuries vs. gold). At every month-end, and inside each module separately:

- **Relative momentum** picks the asset with the higher 12-month total return.
- **Absolute momentum** holds that asset only if its 12-month return beats T-bills; otherwise the module moves to T-bills.

The paper's headline result is the decomposition in its Table 14: **both filters add return, but only absolute momentum substantially reduces volatility and drawdown**. Over 1974–2011 the four-module composite earned 14.90% a year with a maximum drawdown of –10.92%, against an equal-weight basket of the same asset classes, held without momentum, which returned 9.93% with a –27.00% drawdown. Those results are in-sample and before costs, and the 2016 revision did not extend the data beyond 2011.

How this backtest connects to it

The paper's data stop in December 2011, so every month since then is a genuine out-of-sample test. I rebuilt the system with ETFs that track the paper's indices, kept every rule and parameter unchanged, and ran it through a backtesting engine.

Module	Paper indices (asset A / asset B)	ETFs used
Equities	MSCI US / EAFE+ (MSCI ACWI ex US)	SPY / ACWX
Credit	BofA US Cash Pay High Yield / US Intermediate Credit	HYG / IGIB
REITs	FTSE Nareit Equity REITs / Mortgage REITs	USRT / REM
Economic Stress	US Long Treasury / Gold	VGLT / GLD
Safe asset	3-month US T-bills	BIL, then SGOV from Jul 2020

- **Same rules as the paper:** 12-month lookback with no skip month, 25% per module, monthly rebalancing.
- **Realistic execution:** the signal uses the last close of each month and trades at the next session's open. The paper does not specify execution, and trading on the signal's own close would introduce look-ahead bias.
- **Net of costs:** 5 bps on turnover, on total-return prices (dividends reinvested) from FactSet.
- **Benchmark:** the same eight assets plus T-bills, equally weighted (1/9 each) and without momentum, mirroring the paper's own reference.

Key findings

- **The edge reversed.** The strategy returned 4.32% a year against 6.19% for the no-momentum basket (–1.87 pp, versus +4.97 pp in-sample), and it trailed in both halves of the period.
- **Absolute momentum did not protect.** The maximum drawdown was –28.62%, identical to relative momentum alone and deeper than the basket's –24.89%. A monthly signal is too slow to exit a fast crash such as COVID-19 and too slow to re-enter afterwards: 2020 closed at –12.23% for the strategy against +8.49% for the basket.
- **The safe asset stopped paying.** T-bills returned 0.41% a year until mid-2019 and 2.78% afterwards, against close to 6% in the paper's sample, so every exit parked capital at near-zero rates.
- **Relative momentum is the only component with out-of-sample support**, and a modest one: +1.44 pp a year over the basket.

Bottom line: the paper's thesis does not replicate out of sample. Any variant chosen after seeing these results (dropping REITs, changing the filter, shortening the lookback) would be a new in-sample model that needs its own test period.

Out-of-sample vs. the original paper

Metric	Paper (in-sample)	This backtest (out-of-sample)
Period	Jan 1974 – Dec 2011 (38 years)	Dec 2011 – Sep 2026 (14.7 years)
Data	Monthly index returns	Daily ETF prices, total return
Transaction costs	None deducted	5 bps on turnover
Annual return (CAGR)	14.90%	4.32%
Volatility	7.99%	8.96%
Sharpe ratio (rf = T-bills)	1.07	0.38
Maximum drawdown	–10.92%	–28.62%
Positive months	73%	63.6%
Return vs. no-momentum basket	+4.97 pp	–1.87 pp

Effect decomposition (replication of the paper's Table 14)

Variant	Out-of-sample (2012–2026)						Paper (1974–2011)		
	CAGR	Vol.	Sharpe (rf = 0)	Sharpe (rf = T-bills)	Max DD	Positive months	CAGR	Sharpe (rf = T-bills)	Max DD
No momentum	6.19%	9.00%	0.71	0.56	–24.89%	67.0%	9.93%	0.50	–27.00%
Absolute only	3.29%	6.21%	0.55	0.33	–22.27%	65.3%	11.76%	1.05	–7.52%
Relative only	7.63%	10.53%	0.75	0.64	–28.62%	66.5%	14.21%	0.80	–27.29%
Dual momentum	4.32%	8.96%	0.52	0.38	–28.62%	63.6%	14.90%	1.07	–10.92%

Operating profile by module

Out-of-sample figures; the paper's in brackets.

Module	Pair (A / B)	Switches per year	Time in A	Time in B	Time in T-bills
Equities	SPY / ACWX	1.84 (1.4)	71.8% (37.7%)	16.4% (39.7%)	11.9% (22.6%)
Credit	HYG / IGIB	1.29 (1.2)	58.8% (55.3%)	29.4% (19.5%)	11.9% (25.2%)
REITs	USRT / REM	2.65 (1.6)	45.8% (46.9%)	31.1% (26.8%)	23.2% (26.3%)
Economic Stress	VGLT / GLD	2.58 (1.6)	28.2% (43.2%)	46.9% (39.0%)	24.9% (17.8%)

Calendar-year returns

Year	Dual Momentum	No-momentum basket	Difference	Sharpe (rf = 0)
2012	8.10%	10.66%	-2.56 pp	1.42
2013	5.37%	0.17%	+5.20 pp	0.78
2014	8.85%	9.01%	-0.16 pp	1.76
2015	-0.52%	-2.55%	+2.03 pp	-0.03
2016	-0.10%	8.43%	-8.53 pp	0.02
2017	12.35%	11.56%	+0.79 pp	2.52
2018	-6.58%	-3.03%	-3.55 pp	-0.90
2019	12.11%	18.18%	-6.07 pp	2.06
2020	-12.23%	8.49%	-20.72 pp	-0.57
2021	8.12%	9.37%	-1.25 pp	1.03
2022	-9.48%	-15.31%	+5.83 pp	-1.34
2023	0.29%	12.95%	-12.66 pp	0.08
2024	16.47%	8.34%	+8.13 pp	1.73
2025	22.33%	17.09%	+5.24 pp	1.90
2026	4.67%	3.00%	+1.67 pp	0.53

Notes

- Maximum drawdown is the deepest point of the daily equity curve below its previous peak. The paper only has month-end data, which miss intra-month troughs, so its figures are a lower bound of what daily data would show.
- 2026 is a partial year (to 2026-09-25). ETFs approximate the paper's indices (SPY tracks the S&P 500 rather than MSCI US; HYG tracks iBoxx rather than BofA high yield) and their fees are already in the prices. Dividends are reinvested gross of withholding tax.
- Sharpe (rf = 0) uses daily net returns, annualized with $\sqrt{252}$; the calendar-year column applies it to each year's daily returns. Sharpe (rf = T-bills) is measured in excess of T-bills, as in the paper, on monthly returns annualized with $\sqrt{12}$.
- For research and educational purposes only. Past performance, simulated or real, does not guarantee future results. This is not investment advice.
- Code (MIT License): github.com/FranciscoAC-burst/dual-momentum-oos. Price data are not included.